AP CALCULUS AB

Unit 1 • Lesson 1.6 notes

Lesson 1.6 • Determining Limits Using Algebraic Manipulations

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Recognize the indeterminate form $\frac{0}{0}$.
- Rewrite expressions by factoring, using conjugates, and combining fractions to evaluate limits.

Vocabulary. indeterminate form, factoring, conjugate, complex fraction.

1. Why $\frac{0}{0}$ needs work

$\frac{0}{0}$ is not a number: it signals that a common factor can usually be removed. The rewritten expression agrees with the original for $x \neq c$, so it has the same limit.

Example 1 • Factoring

$$\lim_{x\to -3} \frac{x^2+x-6}{x^2-9} = \lim_{x\to -3} \frac{(x+3)(x-2)}{(x+3)(x-3)} = \frac{-5}{-6} = \frac{5}{6}.$$

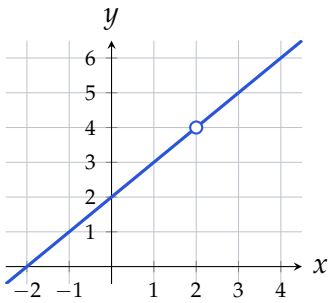
Example 2 • Conjugate

$$\lim_{x\to 9} \frac{\sqrt{x}-3}{x-9} = \lim_{x\to 9} \frac{\sqrt{x}-3}{(\sqrt{x}-3)(\sqrt{x}+3)} = \frac{1}{6}.$$

Example 3 • Complex fraction

$$\lim_{h\to 0} \frac{\frac{1}{x+h}-\frac{1}{x}}{h} = \lim_{h\to 0} \frac{\frac{x-(x+h)}{x(x+h)}}{h} = \lim_{h\to 0} \frac{-1}{x(x+h)} = -\frac{1}{x^2}.$$

Example 4 • Graph check



Solution. The graph of $\dfrac{x^2-4}{x-2}$ is the line $y = x+2$ with a hole at $(2,4)$, confirming $\lim_{x\to 2} \frac{x^2-4}{x-2} = 4$.

Common mistakes

- Concluding that a $\frac{0}{0}$ limit does not exist, or equals 0 or 1.
- Cancelling terms instead of factors: $\frac{x^2-4}{x-2} \neq x-2$.

Quick check

1. $\lim_{x\to 1} \dfrac{x^3-1}{x-1}$ 2. $\lim_{x\to 0} \dfrac{\sqrt{1+x}-1}{x}$

Answers. 1. $x^2+x+1 \to 3$. 2. $\frac{1}{\sqrt{1+x}+1} \to \frac{1}{2}$.